

Instructions

This mock test mirrors the Infosys SP/DSE online assessment structure.

Section 1 - Logical Reasoning: 10 questions (~17 min).

Section 2 - Quantitative Ability: 10 questions (~20 min).

Section 3 - Verbal Ability: 8 questions (~10 min).

Section 4 - Pseudocode / Technical: 8 questions (~20 min).

Section 5 - Coding: 2 problems (~45 min).

Attempt in one sitting. Correct answers are highlighted for review after you finish.

Section 1 - Logical Reasoning

Q1. Find the next term: 2, 6, 12, 20, 30, ?

(A) 36

(B) 40

(C) 42 ✓

(D) 44

Answer: (C) 42

Q2. Find the next term: 1, 4, 9, 16, 25, ?

(A) 30

(B) 36 ✓

(C) 49

(D) 35

Answer: (B) 36

Q3. Find the next term: 3, 6, 12, 24, ?

(A) 36

(B) 48 ✓

(C) 30

(D) 42

Answer: (B) 48

Q4. If letters are replaced by their opposite (A<->Z, B<->Y, ...), then CHAIR is coded as:

(A) XSZRI ✓

(B) XSAIR

(C) XSZIR

(D) XRZSI

Answer: (A) XSZRI

Q5. If FRIEND is coded as HTKGPF (each letter +2), then CANDLE is coded as:

(A) ECPFNG ✓

(B) ECQFNG

(C) DBOEMF

(D) ECPGNF

Answer: (A) ECPFNG

Q6. P is the brother of Q, Q is the sister of R, and R is the father of S. How is P related to S?

(A) Father

(B) Uncle ✓

(C) Brother

(D) Grandfather

Answer: (B) Uncle

Q7. In $P + Q - R$, where 'A + B' means A is the father of B and 'A - B' means A is the sister of B, how is P related to R?

(A) Brother

(B) Uncle

(C) Father ✓

(D) Grandfather

Answer: (C) Father

Q8. A man walks 5 km north, turns right and walks 3 km, then turns right and walks 5 km. How far is he from the start?

(A) 5 km

(B) 3 km ✓

(C) 8 km

(D) 13 km

Answer: (B) 3 km

Q9. A boy walks 4 km east and then 3 km north. How far is he from the starting point?

(A) 7 km

(B) 5 km ✓

(C) 6 km

(D) 1 km

Answer: (B) 5 km

Q10. All cats are dogs. All dogs are animals. Conclusion: All cats are animals.

(A) Follows ✓

(B) Does not follow

(C) Cannot be determined

(D) Partially follows

Answer: (A) Follows

Section 2 - Quantitative Ability

Q1. Find 40% of 60% of 500.

(A) 100

(B) 120 ✓

(C) 150

(D) 200

Answer: (B) 120

Q2. If the price of a commodity increases by 25%, by what percent must consumption be reduced to keep expenditure unchanged?

(A) 25%

(B) 20% ✓

(C) 16.66%

(D) 10%

Answer: (B) 20%

Q3. Cost price is Rs. 400 and selling price is Rs. 500. Find the profit percent.

(A) 20%

(B) 25% ✓

(C) 10%

(D) 15%

Answer: (B) 25%

Q4. An article sold at a 20% loss fetched Rs. 480. Find the cost price.

(A) Rs. 600 ✓

(B) Rs. 576

(C) Rs. 500

(D) Rs. 620

Answer: (A) Rs. 600

Q5. A completes a work in 10 days and B in 15 days. Working together, in how many days will they finish?

(A) 5

(B) 6 ✓

(C) 7.5

(D) 8

Answer: (B) 6

Q6. A is twice as efficient as B and together they finish a work in 12 days. In how many days can A alone finish it?

(A) 18 ✓

(B) 24

(C) 16

(D) 20

Answer: (A) 18

Q7. Convert 90 km/h into m/s.

(A) 25 m/s ✓

(B) 30 m/s

(C) 20 m/s

(D) 15 m/s

Answer: (A) 25 m/s

Q8. A 150 m long train running at 90 km/h crosses a pole in:

(A) 6 s ✓

(B) 8 s

(C) 10 s

(D) 5 s

Answer: (A) 6 s

Q9. Find the unit digit of 7^{95} .

(A) 1

(B) 3 ✓

(C) 7

(D) 9

Answer: (B) 3

Q10. Find the largest number that divides 62, 132 and 237 leaving the same remainder in each case.

(A) 30

(B) 35 ✓

(C) 42

(D) 70

Answer: (B) 35

Section 3 - Verbal Ability

Q1. Choose the synonym of ABUNDANT.

(A) Scarce

(B) Plentiful ✓

(C) Rare

(D) Empty

Answer: (B) Plentiful

Q2. Choose the synonym of CANDID.

(A) Secretive

(B) Frank ✓

(C) Rude

(D) Shy

Answer: (B) Frank

Q3. Choose the antonym of TRANSPARENT.

(A) Clear

(B) Opaque ✓

(C) Bright

(D) Thin

Answer: (B) Opaque

Q4. Choose the antonym of OPTIMIST.

(A) Realist

(B) Pessimist ✓

(C) Idealist

(D) Dreamer

Answer: (B) Pessimist

Q5. Choose the correct sentence (subject-verb agreement, negation):

(A) He do not like coffee.

(B) He does not likes coffee.

(C) He does not like coffee. ✓

(D) He not like coffee.

Answer: (C) He does not like coffee.

Q6. Choose the correct sentence (indefinite pronoun 'one of'):

(A) One of the boys are absent.

(B) One of the boys is absent. ✓

(C) One of the boy is absent.

(D) One of boys are absent.

Answer: (B) One of the boys is absent.

Q7. He is good ____ mathematics.

(A) in

(B) at ✓

(C) on

(D) with

Answer: (B) at

Q8. She has been living here ____ 2010.

(A) for

(B) from

(C) since ✓

(D) by

Answer: (C) since

Section 4 - Pseudocode / Technical

Q1. What is the time complexity of binary search on a sorted array of n elements?

(A) $O(n)$

(B) $O(\log n)$ ✓

(C) $O(n \log n)$

(D) $O(1)$

Answer: (B) $O(\log n)$

Q2. Which data structure works on the FIFO principle?

(A) Stack

(B) Queue ✓

(C) Tree

(D) Graph

Answer: (B) Queue

Q3. Which data structure works on the LIFO principle?

(A) Queue

(B) Stack ✓

(C) Linked List

(D) Heap

Answer: (B) Stack

Q4. What is the worst-case time complexity of bubble sort?

(A) $O(n)$

(B) $O(n \log n)$

(C) $O(n^2)$ ✓

(D) $O(\log n)$

Answer: (C) $O(n^2)$

Q5. Which traversal of a binary search tree yields elements in sorted order?

(A) Preorder

(B) Postorder

(C) Inorder ✓

(D) Level order

Answer: (C) Inorder

Q6. function $f(n)$ { if($n \leq 1$) return 1; return $n * f(n-1)$; } What does $f(4)$ return?

(A) 12

(B) 24 ✓

(C) 16

(D) 6

Answer: (B) 24

Q7. for($i=0$; $i<3$; $i++$) print(i); What is the output?

(A) 1 2 3

(B) 0 1 2 ✓

(C) 0 1 2 3

(D) 0 1

Answer: (B) 0 1 2

Q8. $x=10$; $y=20$; $x=x+y$; $y=x-y$; $x=x-y$; What are the final values of x and y ?

(A) $x=10$, $y=20$

(B) $x=20$, $y=10$ ✓

(C) $x=30$, $y=10$

(D) $x=20$, $y=20$

Answer: (B) $x=20$, $y=10$

Section 5 - Coding

Problem 1: Second Largest Element in an Array

Difficulty: *Easy* | Pattern: *Array Traversal*

Problem Statement

Given an array of N integers, find the second largest element. If it does not exist, print -1. Duplicates of the largest value are ignored when finding the second largest.

Input Format

First line: N . Second line: N space-separated integers.

Output Format

A single integer: the second largest element, or -1.

Constraints

$1 \leq N \leq 10^5$, $-10^9 \leq a[i] \leq 10^9$

Sample Input

```
6
12 35 1 10 34 1
```

Sample Output

```
34
```

Python Solution

```
def second_largest(arr):
    largest = second = float('-inf')
    for x in arr:
        if x > largest:
            second = largest
            largest = x
        elif largest > x > second:
            second = x
    return second if second != float('-inf') else -1

arr = [12, 35, 1, 10, 34, 1]
print(second_largest(arr))
```

Step-by-Step Explanation

Track the largest and second largest in a single pass. When a new maximum appears, the old maximum becomes the second largest. A value strictly between the current second and largest updates the second largest. This avoids sorting and runs in linear time.

Dry Run

```
arr = [12, 35, 1, 10, 34, 1]
12 -> largest=12, second=-inf
35 -> largest=35, second=12
1 -> no change
10 -> second=12 (10<12) no change
34 -> 35>34>12 -> second=34
1 -> no change
Result = 34
```

Time Complexity: **O(N)** | Space Complexity: **O(1)**

Problem 2: Coin Change - Minimum Number of Coins

Difficulty: **Medium** | Pattern: **Dynamic Programming (unbounded knapsack)**

Problem Statement

Given coin denominations and a target amount, find the minimum number of coins needed to make the amount. Each coin can be used unlimited times. If it cannot be formed, print -1.

Input Format

Line 1: number of coin types and the amount. Line 2: the coin denominations.

Output Format

Minimum number of coins, or -1 if impossible.

Constraints

$1 \leq \text{coins} \leq 12$, $1 \leq \text{amount} \leq 10^4$, $1 \leq \text{coin}[i] \leq 10^4$

Sample Input

```
3 11
1 2 5
```

Sample Output

```
3
```

Python Solution

```
def coin_change(coins, amount):
    INF = amount + 1
    dp = [0] + [INF] * amount
    for a in range(1, amount + 1):
        for c in coins:
            if c <= a:
                dp[a] = min(dp[a], dp[a - c] + 1)
    return dp[amount] if dp[amount] != INF else -1

coins = [1, 2, 5]
print(coin_change(coins, 11))
```

Step-by-Step Explanation

dp[a] holds the fewest coins to make amount a. Base case dp[0]=0. For every amount a, try each coin c: if c fits, dp[a] can be improved to dp[a-c]+1. The answer is dp[amount]. 11 = 5 + 5 + 1 uses 3 coins.

Dry Run

```
dp[0]=0
```

```
dp[1]=1 (1)
dp[2]=1 (2)
dp[5]=1 (5)
dp[10]=2 (5+5)
dp[11]=dp[6]+1=3 (5+5+1)
Result = 3
```

Time Complexity: **O(amount * number_of_coins)** | Space Complexity: **O(amount)**

Infosys

Full-Length Mock Test

Applied Role(s):

Digital Specialist Engineer (Trainee) | Specialist Programmer L1 / L2 / L3 (Trainee)

Online Assessment Preparation Material

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Compiled from verified previous-year patterns, community-reported questions and research-backed company-style practice.

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